Activity No: 6.2	
Activity Title: Built-in Functions	
Course Code: CPE0997	**Program:** Computer Engineering
Course Title: Programming Logic and Design	**Date Performed:** 10 / 22 / 2026
Section: CPE11S2	**Date Submitted:** 10 / 22 / 2026
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Objectives

This activity aims to develop programming skills using functions.

2. Intended Learning Outcomes (ILOs):

The students should be able to:

2.1 Understand how to construct programs modularly from small pieces called functions.

2.2 Implement common math functions available in the C standard library

2.3 Create new functions

Instructions

- *Read the problem statement*
- *Write a C program*
- *Test, Debug and execute the C program*
- *Copy and paste the code and screenshot of the output then submit it using doc, docx, and pdf formats only.*

1. Create a program that defines a function to compute for the area of a cube. The formula of the area of the cube is given as A = s * s * s.
2. Define a function **hypotenuse** that calculates the length of the hypotenuse of a right triangle when the other two sides are given. Use this function in a program to determine the length of the hypotenuse for each of the following triangles. The function takes two arguments of type **double** and return the hypotenuse as a **double.**
3. Implement the following integer functions:
    1. Function **celsius** returns the Celsius equivalent of a Fahrenheit temperature.
    2. Function **fahrenheit** returns the Fahrenheit equivalent of a Celsius temperature.
    3. Use these functions to write a program that prints charts showing the Fahrenheit equivalents of a Celsius temperatures from 0 to 100 degrees, and the Celsius equivalents of all Fahrenheit temperatures from 32 to 212 degrees. Print the outputs in a neat tabular format that minimizes the number of lines of output while remaining readable.

Output/Answer

1. Area of the cube

```cpp
#include <iostream>
#include <iomanip>

using namespace std;

// Function prototype: tells the compiler cubeArea takes one double
// and returns a double.
double cubeArea(double);

int main()
{
    double side;      // length of one side of the cube
    double area;      // result returned by the function

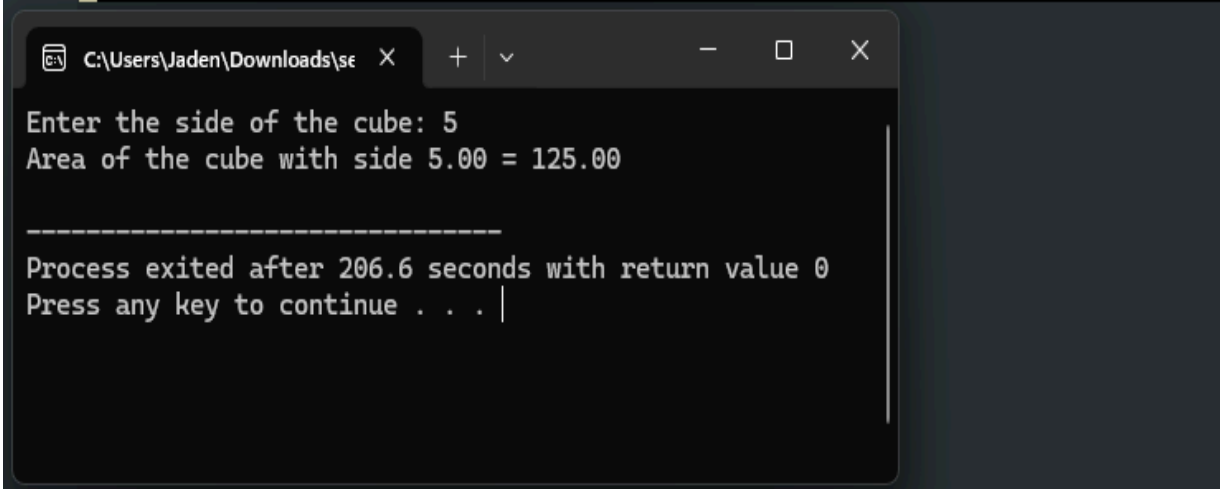
    cout << "Enter the side of the cube: ";
    cin >> side;

    // Function call: 'side' is the argument passed to the function.
    area = cubeArea(side);

    cout << fixed << setprecision(2);
    cout << "Area of the cube with side " << side << " = " << area << endl;

    return 0;
}

// Function definition: 's' is a local variable (parameter) known only here.
double cubeArea(double s)
{
    return s * s * s;    // A = s * s * s
}
```



2. **Hypotenuse**

```cpp
#include <iostream>
#include <iomanip>
#include <cmath>

using namespace std;


double hypotenuse(double, double);

int main()
{
    // The three right triangles to test (side1, side2).
    double side1[3] = {3.0, 5.0, 8.0};
    double side2[3] = {4.0, 12.0, 15.0};

    cout << fixed << setprecision(1);
    cout << setw(10) << "Triangle" << setw(10) << "Side 1"
         << setw(10) << "Side 2" << setw(15) << "Hypotenuse" << endl;

    // Call the function once for each triangle.
    for (int i = 0; i < 3; i++)
    {
        cout << setw(10) << (i + 1)
             << setw(10) << side1[i]
             << setw(10) << side2[i]
             << setw(15) << hypotenuse(side1[i], side2[i]) << endl;
    }

    return 0;
}

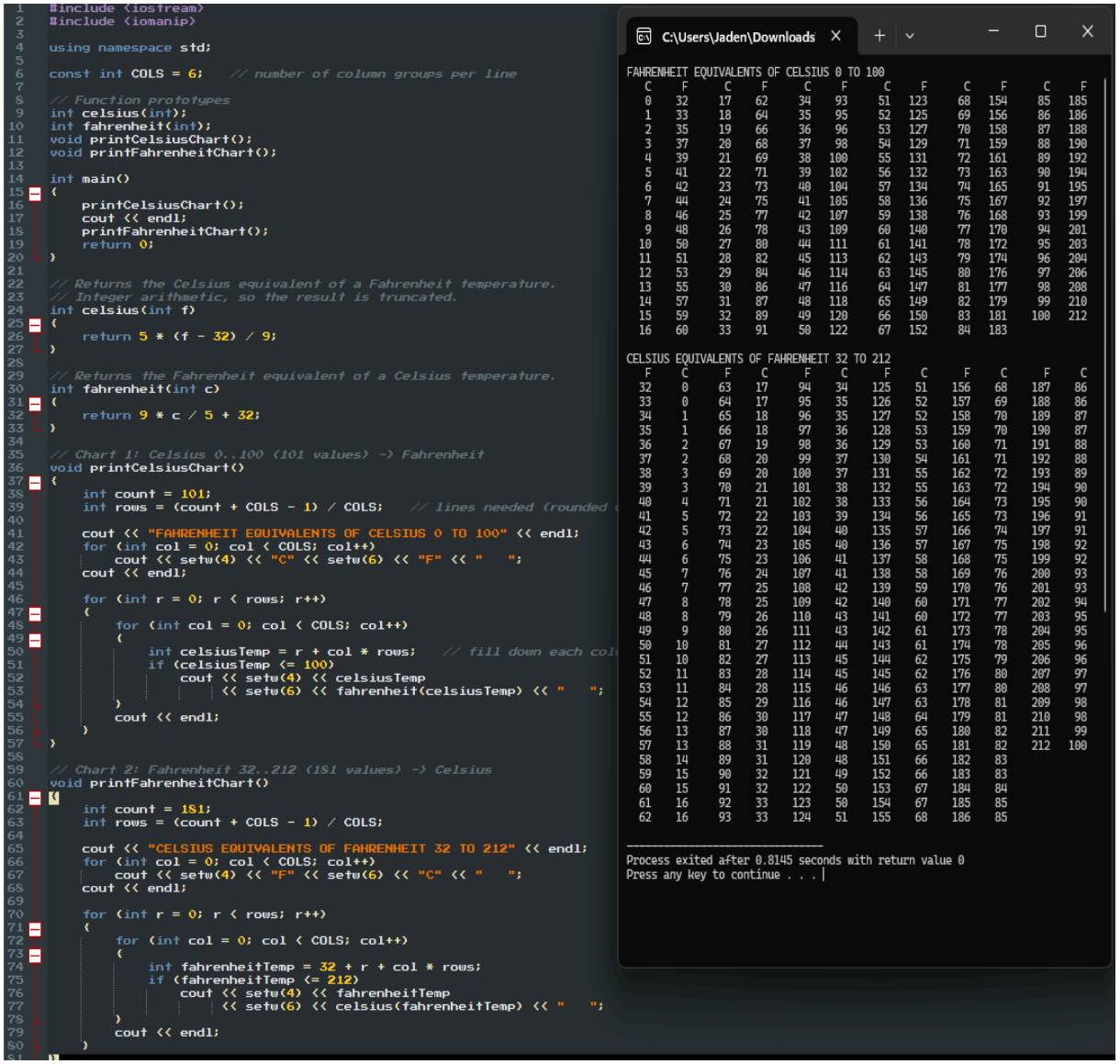
// Pythagorean theorem: c = sqrt(a^2 + b^2)
double hypotenuse(double a, double b)
{
    return sqrt(pow(a, 2) + pow(b, 2));    // pow and sqrt from <cmath>
}
```

```
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Enter the side of the cube: 5
Area of the cube with side 5.00 = 125.00

--------------------------------
Process exited after 206.6 seconds with return value 0
Press any key to continue . . .
```

3. **Temperature**



```cpp
#include <iostream>
#include <iomanip>

using namespace std;

const int COLS = 6;    // number of column groups per line

// Function prototypes
int celsius(int);
int fahrenheit(int);
void printCelsiusChart();
void printFahrenheitChart();

int main()
{
    printCelsiusChart();
    cout << endl;
    printFahrenheitChart();
    return 0;
}

// Returns the Celsius equivalent of a Fahrenheit temperature.
// Integer arithmetic, so the result is truncated.
int celsius(int f)
{
    return 5 * (f - 32) / 9;
}

// Returns the Fahrenheit equivalent of a Celsius temperature.
int fahrenheit(int c)
{
    return 9 * c / 5 + 32;
}

// Chart 1: Celsius 0..100 (101 values) -> Fahrenheit
void printCelsiusChart()
{
    int count = 101;
    int rows = (count + COLS - 1) / COLS;    // lines needed (rounded

    cout << "FAHRENHEIT EQUIVALENTS OF CELSIUS 0 TO 100" << endl;
    for (int col = 0; col < COLS; col++)
        cout << setw(4) << "C" << setw(6) << "F" << "    ";
    cout << endl;

    for (int r = 0; r < rows; r++)
    {
        for (int col = 0; col < COLS; col++)
        {
            int celsiusTemp = r + col * rows;    // fill down each col
            if (celsiusTemp <= 100)
                cout << setw(4) << celsiusTemp
                     << setw(6) << fahrenheit(celsiusTemp) << "    ";
        }
        cout << endl;
    }
}

// Chart 2: Fahrenheit 32..212 (181 values) -> Celsius
void printFahrenheitChart()
{
    int count = 181;
    int rows = (count + COLS - 1) / COLS;

    cout << "CELSIUS EQUIVALENTS OF FAHRENHEIT 32 TO 212" << endl;
    for (int col = 0; col < COLS; col++)
        cout << setw(4) << "F" << setw(6) << "C" << "    ";
    cout << endl;

    for (int r = 0; r < rows; r++)
    {
        for (int col = 0; col < COLS; col++)
        {
            int fahrenheitTemp = 32 + r + col * rows;
            if (fahrenheitTemp <= 212)
                cout << setw(4) << fahrenheitTemp
                     << setw(6) << celsius(fahrenheitTemp) << "    ";
        }
        cout << endl;
    }
}
```

```
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FAHRENHEIT EQUIVALENTS OF CELSIUS 0 TO 100
   C    F      C    F      C    F      C    F      C    F      C    F
   0   32     17   62     34   93     51  123     68  154     85  185
   1   33     18   64     35   95     52  125     69  156     86  186
   2   35     19   66     36   96     53  127     70  158     87  188
   3   37     20   68     37   98     54  129     71  159     88  190
   4   39     21   69     38  100     55  131     72  161     89  192
   5   41     22   71     39  102     56  132     73  163     90  194
   6   42     23   73     40  104     57  134     74  165     91  195
   7   44     24   75     41  105     58  136     75  167     92  197
   8   46     25   77     42  107     59  138     76  168     93  199
   9   48     26   78     43  109     60  140     77  170     94  201
  10   50     27   80     44  111     61  141     78  172     95  203
  11   51     28   82     45  113     62  143     79  174     96  204
  12   53     29   84     46  114     63  145     80  176     97  206
  13   55     30   86     47  116     64  147     81  177     98  208
  14   57     31   87     48  118     65  149     82  179     99  210
  15   59     32   89     49  120     66  150     83  181    100  212
  16   60     33   91     50  122     67  152     84  183

CELSIUS EQUIVALENTS OF FAHRENHEIT 32 TO 212
   F    C      F    C      F    C      F    C      F    C      F    C
  32    0     63   17     94   34    125   51    156   68    187   86
  33    0     64   17     95   35    126   52    157   69    188   86
  34    1     65   18     96   35    127   52    158   70    189   87
  35    1     66   18     97   36    128   53    159   70    190   87
  36    2     67   19     98   36    129   53    160   71    191   88
  37    2     68   20     99   37    130   54    161   71    192   88
  38    3     69   20    100   37    131   55    162   72    193   89
  39    3     70   21    101   38    132   55    163   72    194   90
  40    4     71   21    102   38    133   56    164   73    195   90
  41    5     72   22    103   39    134   56    165   73    196   91
  42    5     73   22    104   40    135   57    166   74    197   91
  43    6     74   23    105   40    136   57    167   75    198   92
  44    6     75   23    106   41    137   58    168   75    199   92
  45    7     76   24    107   41    138   58    169   76    200   93
  46    7     77   25    108   42    139   59    170   76    201   93
  47    8     78   25    109   42    140   60    171   77    202   94
  48    8     79   26    110   43    141   60    172   77    203   95
  49    9     80   26    111   43    142   61    173   78    204   95
  50   10     81   27    112   44    143   61    174   78    205   96
  51   10     82   27    113   45    144   62    175   79    206   96
  52   11     83   28    114   45    145   62    176   80    207   97
  53   11     84   28    115   46    146   63    177   80    208   97
  54   12     85   29    116   46    147   63    178   81    209   98
  55   12     86   30    117   47    148   64    179   81    210   98
  56   13     87   30    118   47    149   65    180   82    211   99
  57   13     88   31    119   48    150   65    181   82    212  100
  58   14     89   31    120   48    151   66    182   83
  59   15     90   32    121   49    152   66    183   83
  60   15     91   32    122   50    153   67    184   84
  61   16     92   33    123   50    154   67    185   85
  62   16     93   33    124   51    155   68    186   85

Process exited after 0.8145 seconds with return value 0
Press any key to continue . . .
```

Supplementary Activity (if any)

Conclusion

- Through these activities, I learned that breaking a program into small functions, like cube area, hypotenuse, celsius and fahrenheit, makes my code easier to write, test and understand than putting everything in main(). I also learned how function prototypes, parameters and return values let different parts of a program pass information to each other while at the same time, keeping their variables separate. Using math library functions, something I haven't tried before like sqrt and pow showed me that I don't have to reinvent common calculations, because I can combine existing tools with my own code. Earlier,

working with files taught me to read data line by line and handle bad input without crashing, and I now see that careful, modular programming is a skill I'll need as a computer engineering student.

Assessment Rubric:

Rubric for SO 7

Criteria	Ratings						Pts
SO 7 PI 1 IILO4 Utilize lifelong learning skills in pursuit of personal development and excellence in professional practice. threshold: 4.8 pts	6 pts Excellent | Educational interests and pursuits exist and flourish outside classroom requirements,knowledge and/or experiences are pursued independently and applies knowledge learned into practice	5 pts Good | Educational interests and pursuits exist and flourish outside classroom requirements,knowledge and/or experiences are pursued independently	4 pts Satisfactory | Look beyond classroom requirements, showing interest in pursuing knowledge independently	3 pts Unsatisfactory | Begins to look beyond classroom requirements, showing interest in pursuing knowledge independently	2 pts Poor | Relies on classroom instruction only	1 pts Very Poor | No initiative or interest in acquiring new knowledge	6 pts
SO 7 PI 2 IILO4 Utilize lifelong learning skills in pursuit of personal development and excellence in professional practice. threshold: 4.8 pts	6 pts Excellent | Completes an assigned task independently and practices continuous improvement	5 pts Good | Completes an assigned task without supervision or guidance	4 pts Satisfactory | Requires minimal guidance to complete an assigned task	3 pts Unsatisfactory | Requires detailed or step-by-step instructions to complete a task	2 pts Poor | Shows little interest to complete a task independently	1 pts Very Poor | No interest to complete a task independently	6 pts
SO 7 PI 3 IILO4 Utilize lifelong learning skills in pursuit of personal development and excellence in professional practice. threshold: 4.8 pts	6 pts Excellent | Synthesizes and integrates information from a variety of sources; formulates a clear and precise perspective; draws appropriate conclusions	5 pts Good | Evaluate information from a variety of sources; formulates a clear and precise perspective.	4 pts Satisfactory | Analyze information from a variety of sources; formulates a clear and precise perspective.	3 pts Unsatisfactory | Apply the gathered information to formulate the problem	2 pts Poor | Gather and summarized the information from a variety of sources but failed to formulate the problem	1 pts Very Poor | Gather information from a variety of sources	6 pts
SO 7 PI 4 IILO4 Utilize lifelong learning skills in pursuit of personal development and excellence in professional practice. threshold: 4.8 pts	6 pts Excellent | Ideas are combined in original and creative ways in line with the new and emerging technology trends to solve a problem or address an issue.	5 pts Good | Ideas are creative and adapt the new knowledge to solve a problem or address an issue	4 pts Satisfactory | Ideas are creative in solving a problem, or address an issue	3 pts Unsatisfactory | Shows some creative ways to solve the problem	2 pts Poor | Shows initiative and attempt to develop creative ideas to solve the problem	1 pts Very Poor | Ideas are copied or restated from the sources consulted	6 pts

Total Points: 24